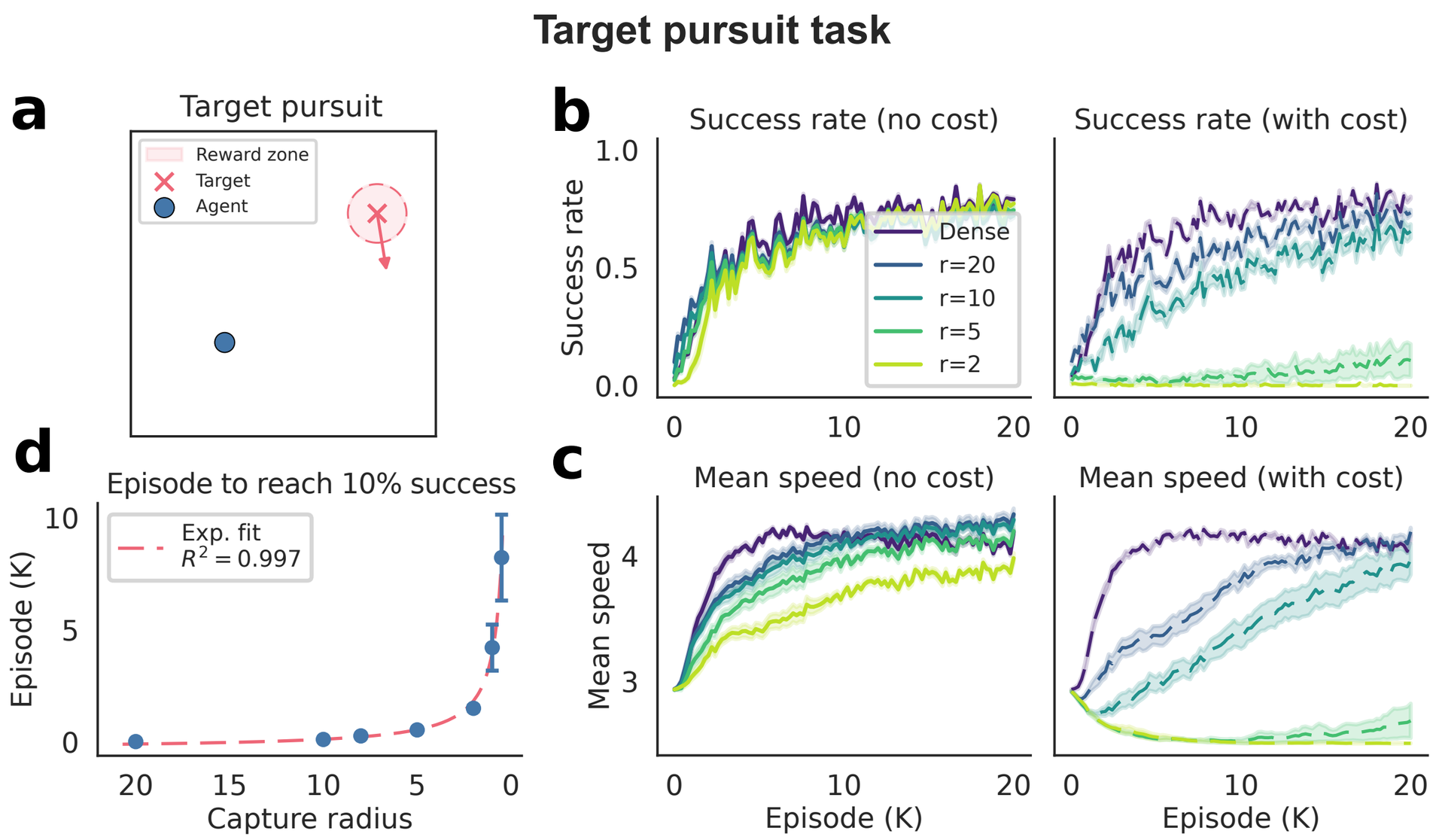

Figure 1: Reward sparsity leads to exponentially slow learning, and produces quitting in the presence of competing optima in the target pursuit task.
a) Schematic of the target pursuit environment. The agent moves in two dimensions and must learn to remain close to the moving target.
b) Success rate over training for various capture radii r (smaller r = sparser reward) and a dense reward baseline. Without an action cost (left), all sparsity levels eventually learn the task. With an action cost (right), sparser rewards lead to slower learning and quitting (r=2).
c) Mean action norm over training. Quitting manifests as a drop in mean speed compared to at initialization and is observed only in the presence of action cost, confirming that competing optima between minimizing effort and pursuing reward drive the quitting behavior.
d) Number of episodes to reach 10% success as a function of capture radius (x-axis reversed so that reward sparsity increases to the right). Learning speed slows exponentially with increasing sparsity, consistent with the theoretical prediction that sparse rewards yield exponentially slow convergence. Shaded area and error bars indicate standard error across 10 seeds.

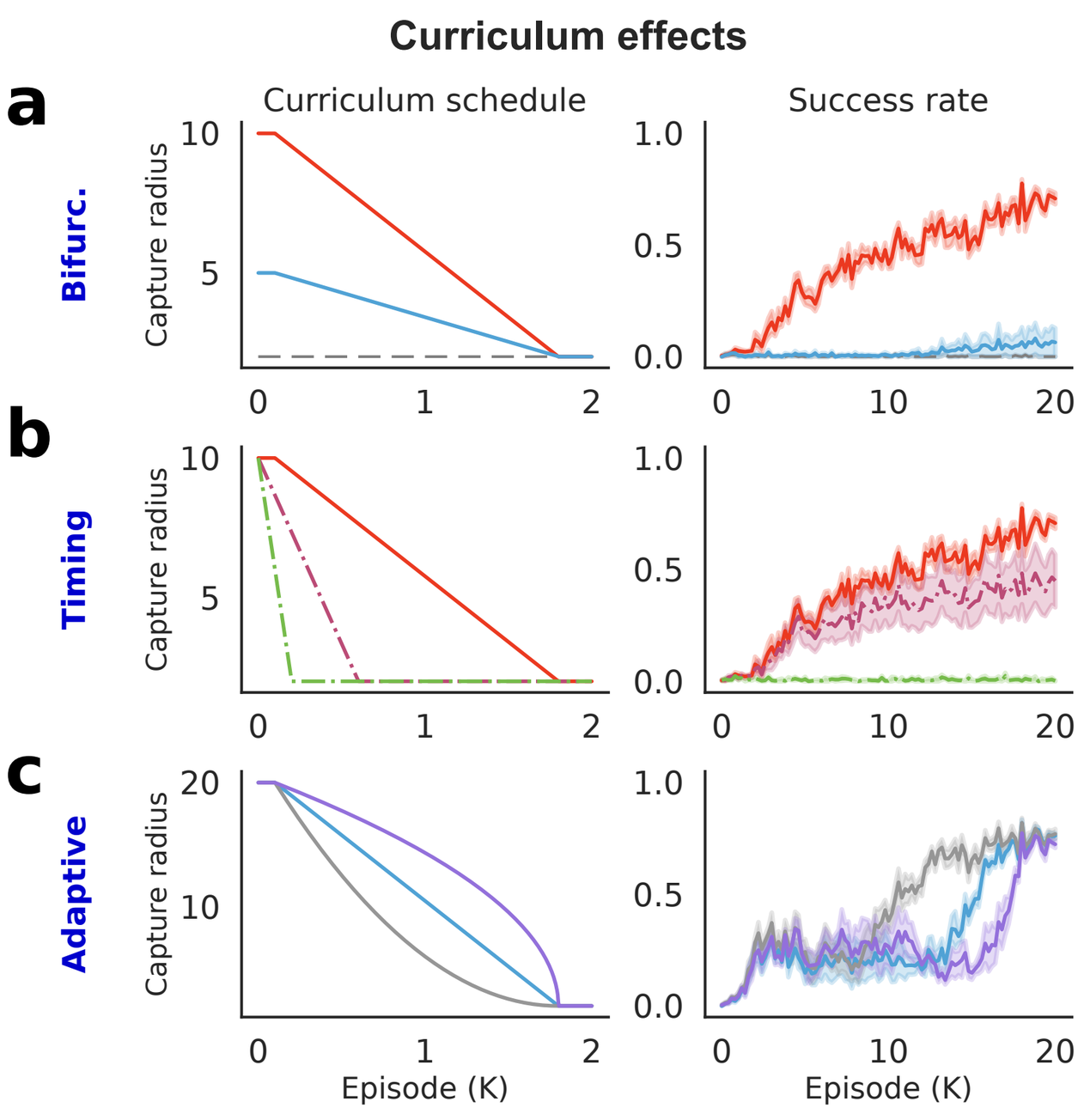

Figure 2: Insights about useful curricula in target pursuit. All runs have a desired final capture radius r=2 and include the action cost. Without a curriculum, r=2 consistently leads to the quitting solution (gray dotted line).
a) Bifurc.: Starting from a larger capture radius r=10 and linearly decreasing to r=2 rescues the agent from quitting. A denser initial reward (red) has much stronger effect on learning than a relatively sparser starting reward (blue).
b) Timing: The timing of reward sparsity transitions matters--transitioning too quickly fails to prevent quitting (green), while a gradual schedule allows the agent to consolidate learning before the task becomes harder.
c) Adaptive: Nonlinear "adaptive" schedules can outperform linear schedules.
While the full bifurcation structure of the loss landscape in these high-dimensional, nonlinear settings is beyond the reach of our analytical framework, **the same factors our theory identifies as critical for curriculum design (initial reward sparsity, transition timing, and schedule shape) still govern learning outcomes here, and can rescue agents from quitting** in the presence of competing optima.

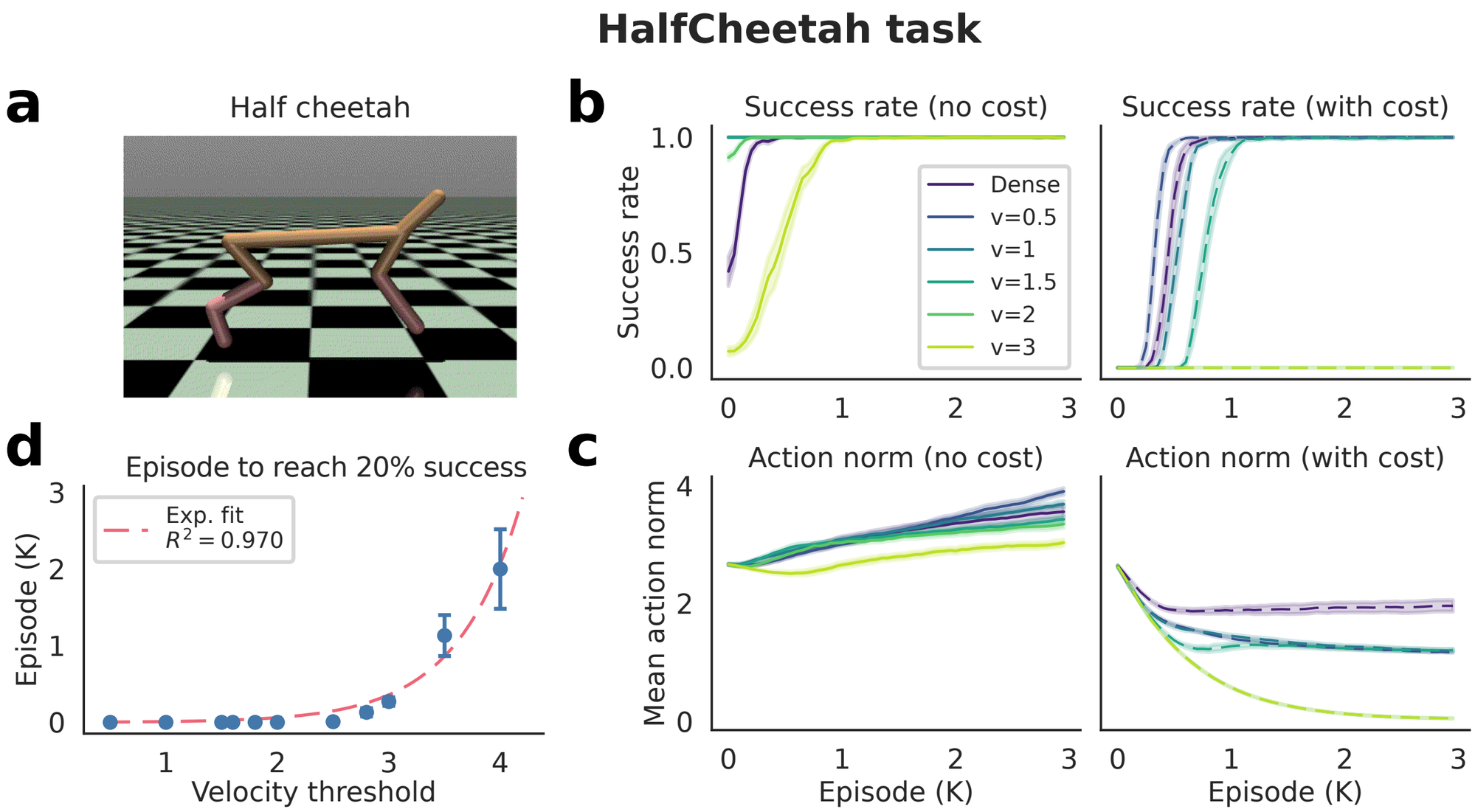

Figure 3: Reward sparsity leads to exponentially slow learning, and produces quitting in the presence of competing optima in the HalfCheetah locomotion task.
a) The HalfCheetah locomotion task: the agent must learn to run forward by coordinating six actuated joints.
b) Success rate (fraction of episodes where forward velocity exceeds threshold v) over training for various velocity thresholds (larger v = sparser reward). Without an action cost (left), all sparsity levels eventually learn. With an action cost (right), sparser rewards cause slower learning and quitting at the most extreme sparsity level (v =3).
c) Mean action norm over training. Quitting manifests as reduced action magnitude and occurs only in the with-cost setting, confirming that competing optima between effort minimization and reward pursuit drive the phenomenon.
d) Episodes to reach 20% success as a function of velocity threshold. Learning slows exponentially with increasing sparsity, replicating the theoretical prediction in a high-dimensional continuous control benchmark. Shaded area and error bars indicate standard error across 10 seeds.

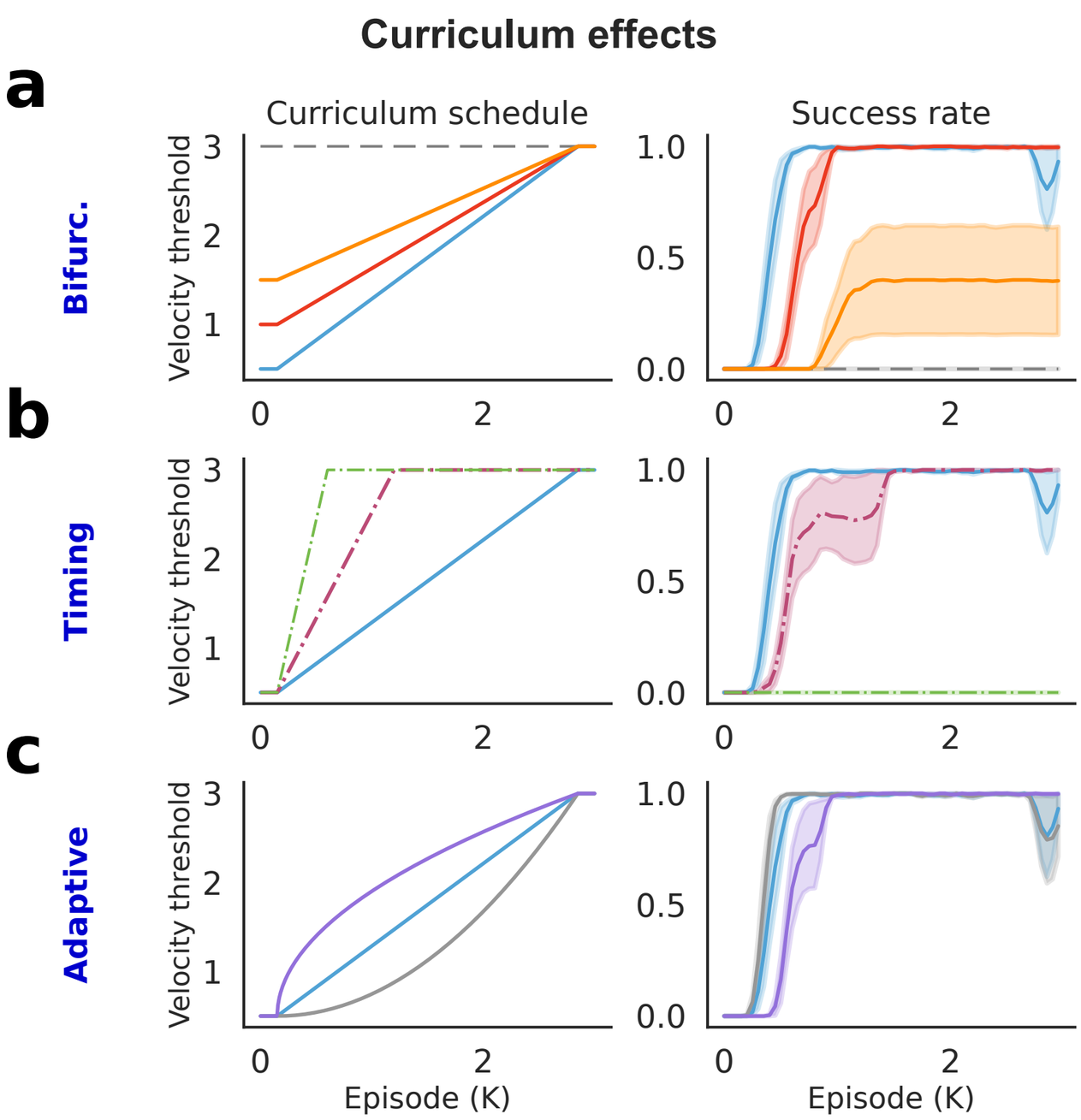

Figure 4: Insights about useful curricula in HalfCheetah.
All runs have a desired final velocity threshold of v=3 and include the action cost penalty. Without a curriculum, all runs with v=3 (grey dotted line) consistently quit.
a) Bifurc.: Starting from an easier velocity threshold (lower v) and transitioning gradually rescues the agents from quitting, with denser initial reward leading to the strongest benefit.
b) Timing: The timing of reward sparsity transitions matters--transitioning too quickly fails to prevent quitting (green), while a gradual schedule allows the agent to consolidate learning before the task becomes harder.
c) Adaptive: Nonlinear "adaptive" schedules can outperform linear schedules.
Shaded area and error bars indicate standard error across 10 seeds.

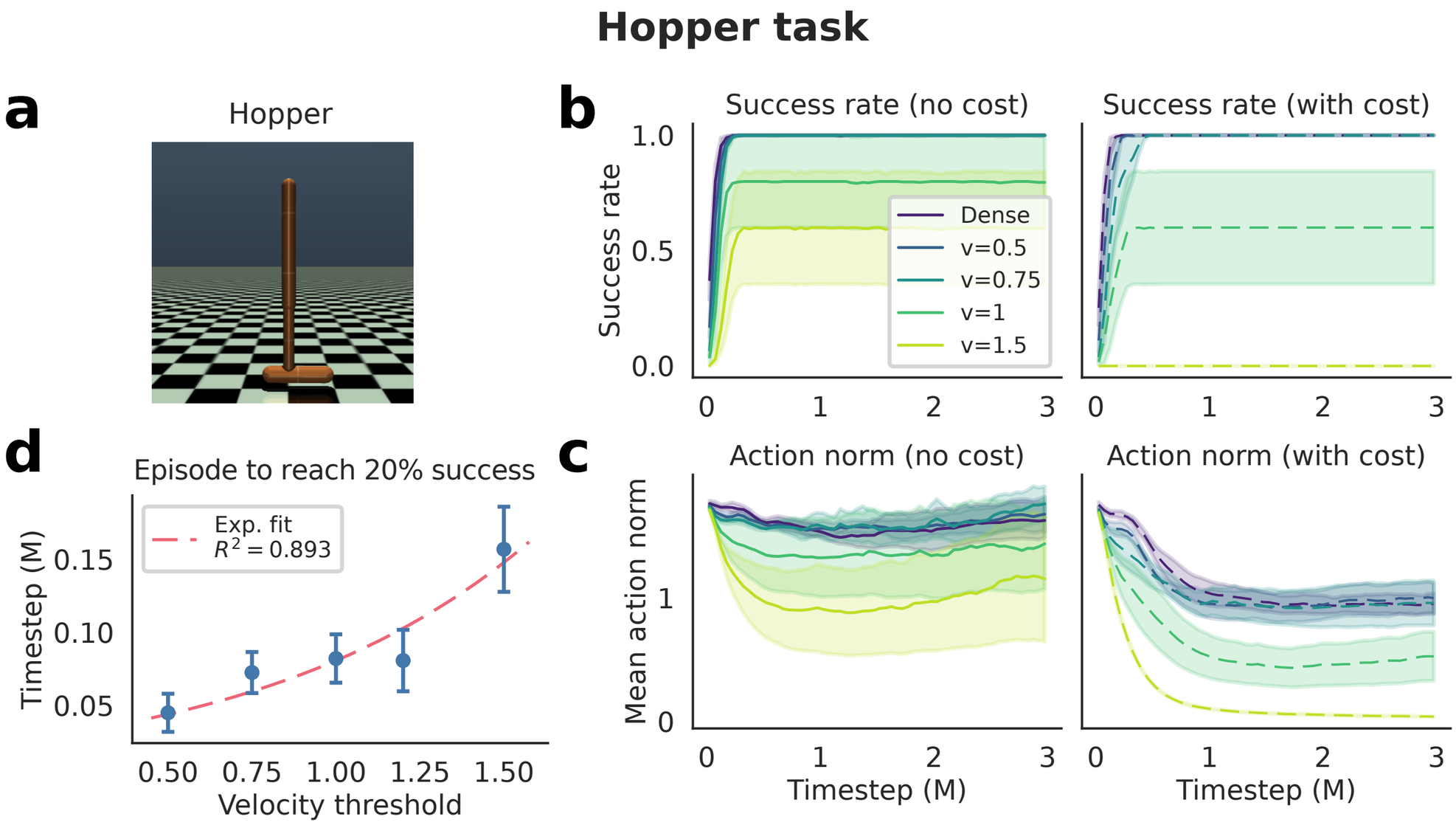

Figure 5: Reward sparsity leads to exponentially slow learning, and produces quitting in the presence of competing optima in the Hopper locomotion task.
a) The Hopper locomotion task: a one-legged agent must learn to hop forward while maintaining balance.
b) Success rate (fraction of episodes where forward velocity exceeds threshold v) over training for various velocity thresholds. Without an action cost (left), all sparsity levels learn, though sparser conditions are slower. With an action cost (right), the sparsest conditions (v = 1.5) exhibit quitting.
c) Mean action norm over training. In the with-cost setting, sparser reward conditions show reduced action norms, indicating quitting behavior where the agent minimizes effort rather than pursuing the locomotion objective.
d) Timesteps to reach 20% success as a function of velocity threshold. Learning slows exponentially with increasing sparsity. Shaded area and error bars indicate standard error across 10 seeds.

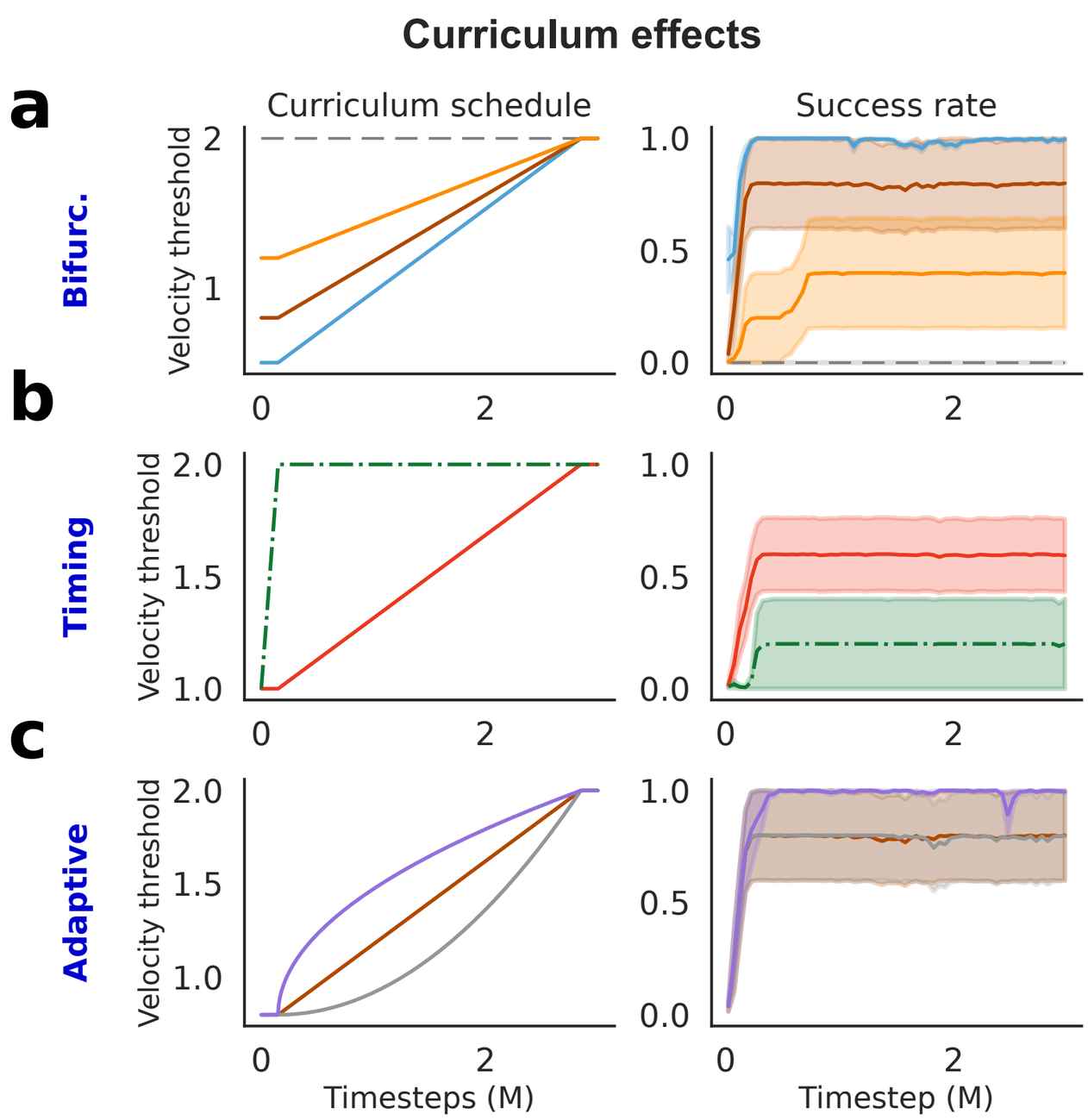

Figure 6: Insights about useful curricula in Hopper. All runs have a desired final velocity threshold of v=2 and include the action cost penalty. Without a curriculum, all runs with v=2 (grey dotted line) consistently quit.
a) Bifurc.: Starting from an easier velocity threshold (lower v) and transitioning gradually rescues the agents from quitting, with denser initial reward leading to the strongest benefit.
b) Timing: The timing of reward sparsity transitions matters--transitioning too quickly fails to prevent quitting (green), while a gradual schedule allows the agent to consolidate learning before the task becomes harder.
c) Adaptive: Nonlinear "adaptive" schedules can outperform linear schedules.
Shaded area and error bars indicate standard error across 10 seeds.